

Processing Pipelines

APPN Aerial SOP — Locked revision 1.0

Processing Pipelines

Version 0.1 – DRAFT, May 2026

Important

This page documents the **standardised GRYFN processing pipelines** used within APPN for UAV-based data processing, intended for trained APPN staff processing CALViS, GOBI, HiRes, and M3M datasets. Adherence to the pipelines below ensures reproducibility, quality assurance, and cross-project comparability. For any processing run that **deviates from the standard pipeline**, detailed records must be kept of every parameter changed, the rationale, and any anticipated implications for data quality.

This page documents the standardised GRYFN processing pipeline YAML files used within APPN for UAV-based data processing. These YAML configurations define consistent, transparent, and repeatable processing workflows across GRYFN-supported sensors and platforms, supporting reproducibility, quality assurance, and cross-project comparability. The files are intended to be used as reference and default templates for approved processing pipelines, with any deviations explicitly documented to maintain traceability and data integrity.

The standard pipelines and the data products they output are detailed below. For a tabular summary of output formats, resolutions and software compatibility, see Standard Data Products.

For additional information about all steps in this document, see the Gryfn Documentation

Important

Document status — work in progress. This page is a draft and requires further revision before it can be considered final.

Outstanding TODOs:

- ☐ CALViS walkthrough author / reviewer: Richard Harwood.
- ☐ Add the GOBI standard processing walkthrough (currently outputs only).
- ☐ Automate handling of GNSS .T04 files in the raw-data formatting step.

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CALViS — Standard Processing Pipeline

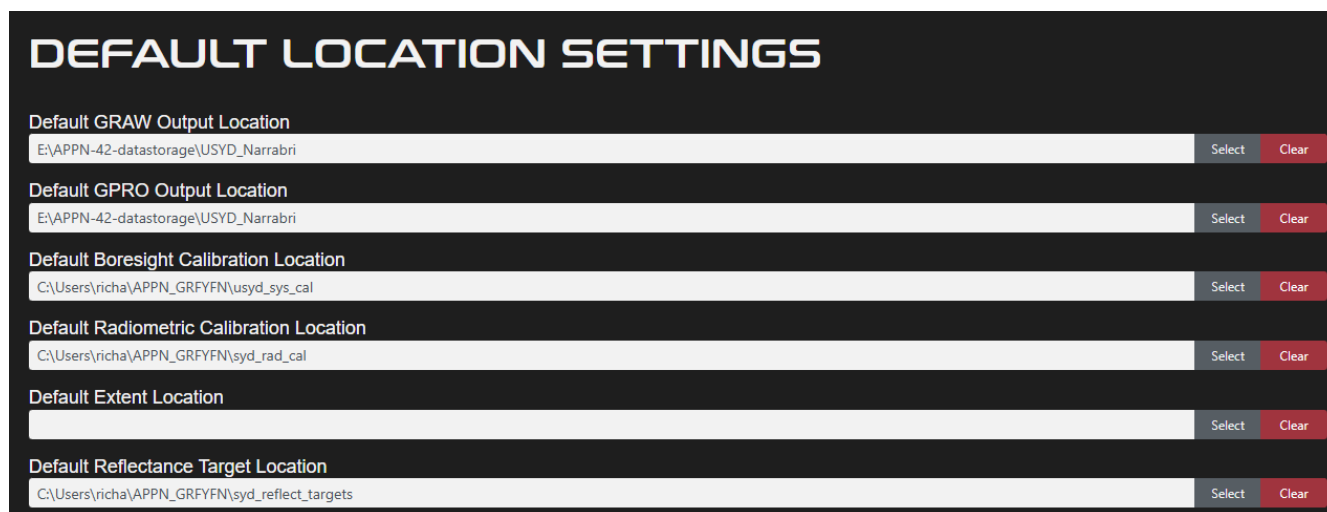
This section describes the standard CALViS processing workflow using the `Calvis_standard_pipeline_v1.0.yaml` file in the GRYFN Processing Tool (**GPT, v1.9.2**).

Note

The pipeline YAML lives in `Protocols/Pipelines/ProcessingPipelines/yaml/`. Download the latest version from the link above before importing it into GPT (see Step 4).

1. Configure GPT defaults

Before processing any data, configure GPT so that it uses the correct radiometric calibration files and points to the correct panel target files.



Setting	Value	Select	Clear
Default GRAW Output Location	E:\APPN-42-datastorage\USYD_Narrabri		
Default GPRO Output Location	E:\APPN-42-datastorage\USYD_Narrabri		
Default Boresight Calibration Location	C:\Users\yricha\APPN_GRFYFN\usyd_sys_cal		
Default Radiometric Calibration Location	C:\Users\yricha\APPN_GRFYFN\syd_rad_cal		
Default Extent Location			
Default Reflectance Target Location	C:\Users\yricha\APPN_GRFYFN\syd_reflect_targets		

Sensor identification (boresight calibration) Each CALViS unit has a unique sensor identifier embedded in the boresight calibration filename. The naming convention is:

Example: `20250527_cAHP-191_SystemCal.yaml`

Where 191 is the USYD sensor number; each CALViS unit is assigned its own unique number.

Important

When processing data, you **must** verify that the sensor number in the boresight calibration file matches the sensor used for data collection.

Radiometric calibration files Each sensor has an associated set of radiometric calibration files supplied by GRYFN.

Important

Verify that the **Radiometric Calibration Location** parameter in GPT points to the correct calibration files for your specific sensor(s).

Reflectance target values The Empirical Line Method (ELM) requires accurate reflectance target values for the calibration panels.

- Four (4) calibration panels are used for the ELM process.
- Each panel has specific, measured target reflectance values.
- These values are unique to your panel set.

Important

Ensure the **Reflectance Target Location** parameter references the correct target values that correspond to your specific calibration panels.

2. Format the raw data

The example CALViS flight used here is **run_00**, a flight from UOA.

A complete CALViS dataset has four key components — **VNIR, SWIR, LiDAR, and GNSS** — and these must be arranged correctly before a working grow can be bundled. The raw data folder will look like this:

2026_APEX > 2026Roseworthy-SA > CALVIS > 20260415 > run_00 > T0_raw >				
Name	Date modified	Type	Size	
100060_20260415_APEX1_AU_dark_2026_04_15_01_23_50	5/6/2026 3:42 PM	File folder		SWIR
100061_20260415_APEX1_AU_2026_04_15_01_24_28	5/6/2026 3:42 PM	File folder		SWIR
20260415_APEX1_AU_2026_04_15_01_24_11_297	5/6/2026 3:47 PM	File folder		VNIR
GNSS	5/6/2026 3:43 PM	File folder		GNSS
lidar	5/6/2026 3:47 PM	File folder		Lidar
Vault	5/6/2026 3:45 PM	File folder		

Note

When you download the VNIR data it has the **LiDAR data nested inside** the VNIR folder, and the **VNIR dark frames** are in the same folder. The **SWIR dark frames** are in a separate folder. The GNSS folder contains the relevant .T04 files (automated handling is WIP).

3. Create the grow bundle

Pre-bundle checks Before each grow, confirm you have:

- ☐ System calibration file (matching the sensor used).
- ☐ Radiometric calibration files.
- ☐ Reflectance target file.

DEFAULT LOCATION SETTINGS

Default GRAW Output Location
 Select Clear

Default GPRO Output Location
 Select Clear

Default Boresight Calibration Location
 Select Clear

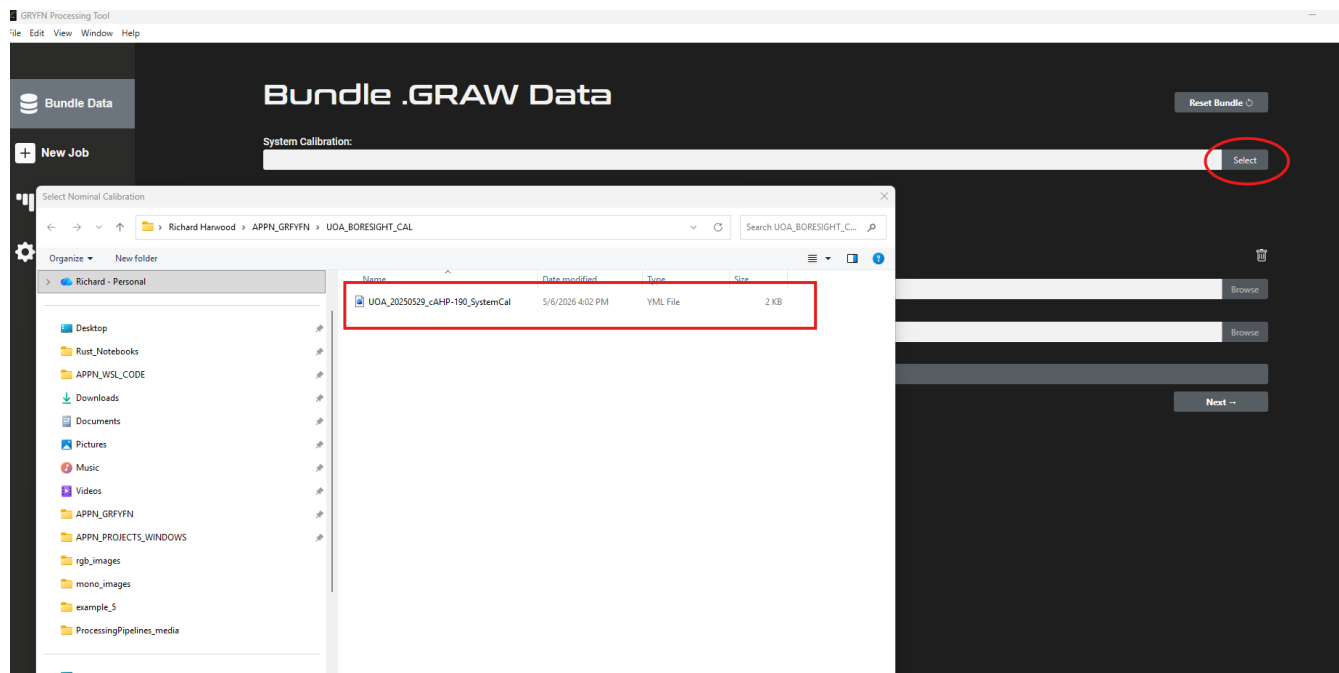
Default Radiometric Calibration Location
 Select Clear

Default Extent Location
 Select Clear

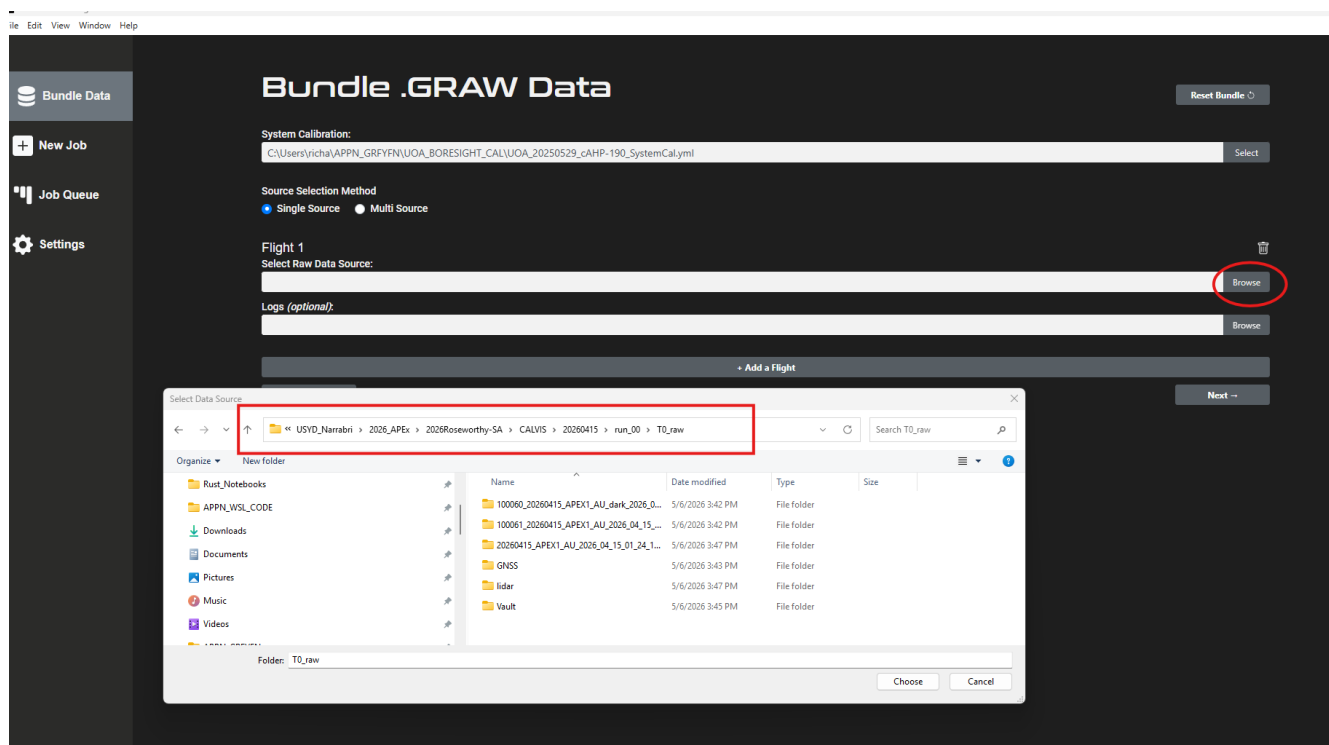
Default Reflectance Target Location
 Select Clear

Bundling steps

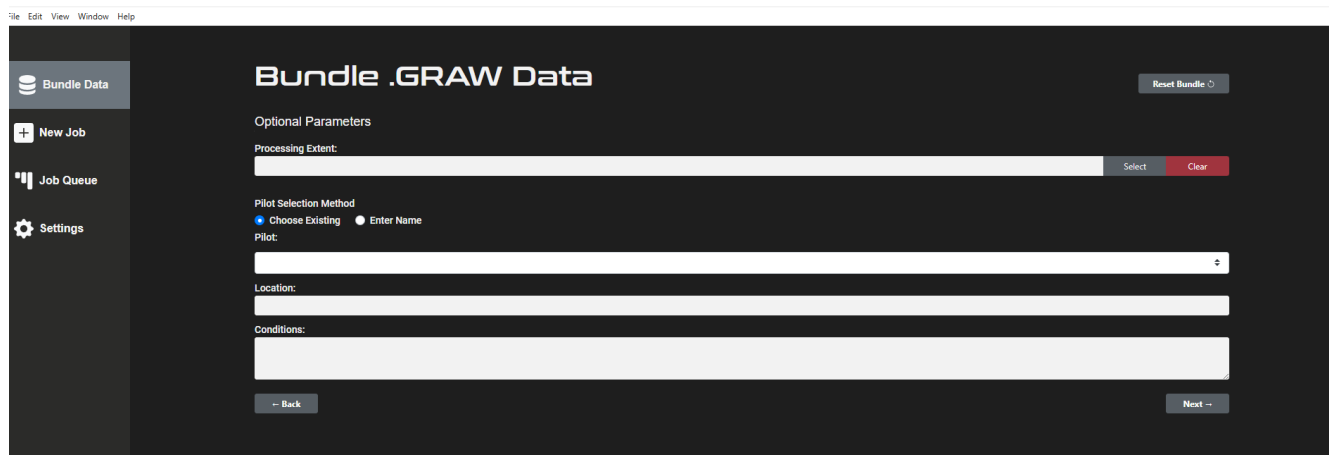
1. Set the system calibration file to match the sensor you are using.



2. Choose the raw data path.



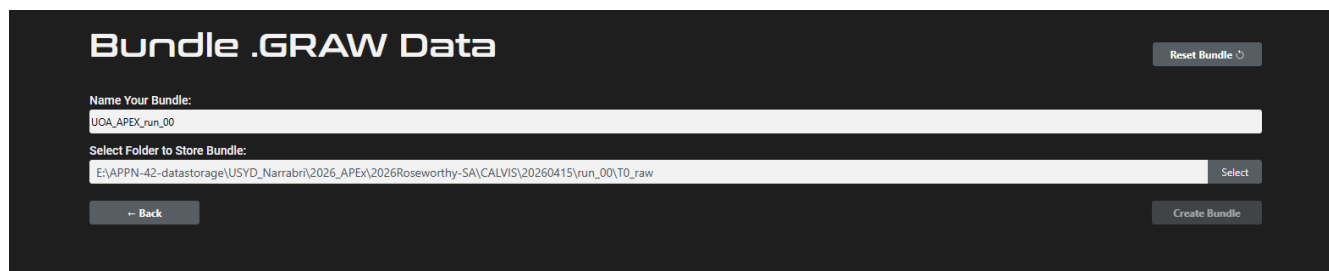
3. Click *Next* to view the optional parameters.



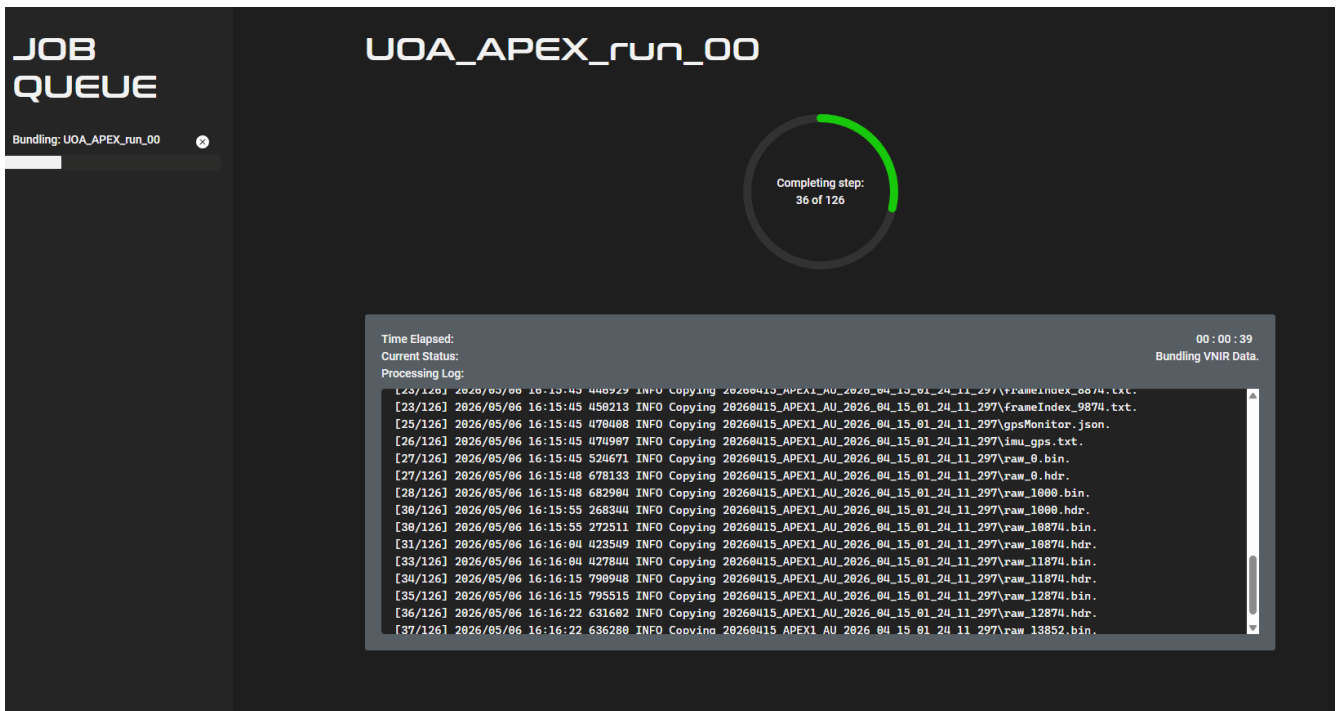
Note

Unless you have a specific reason to set an extent, **leave it blank** — data can be cropped to the hyperspectral capture area later, which is ideal for most standard flights. Other fields here are also optional.

4. Click *Next*. Give the bundle a logical name and save it under T0_raw.



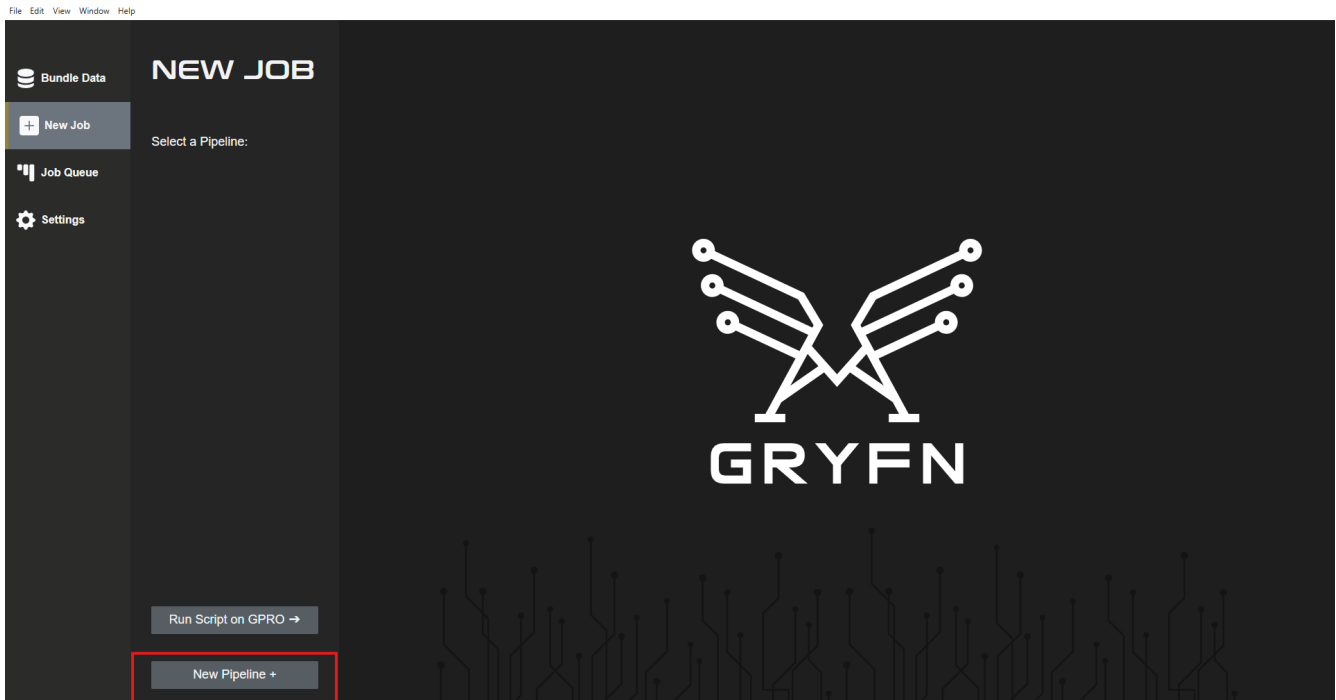
5. Click *Create bundle*. You should see a progress report while the bundle is being created.



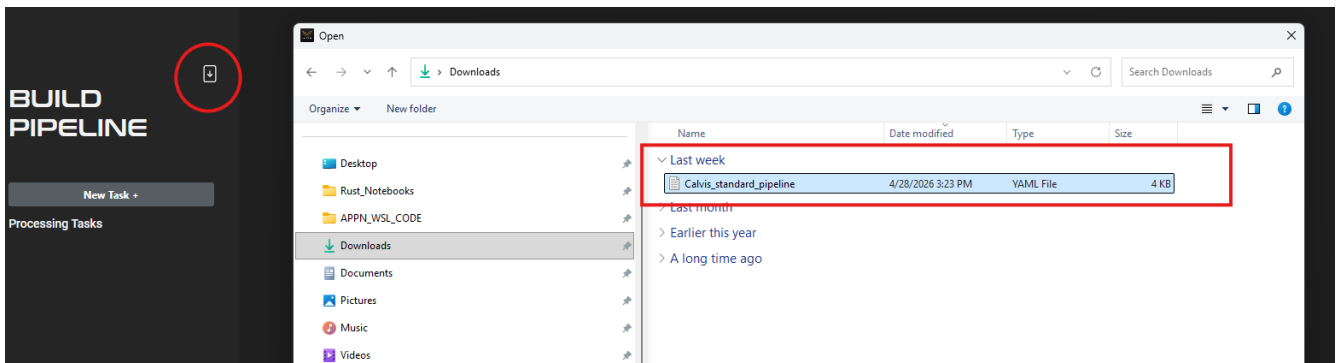
4. Load the Calvis_standard_pipeline

Import the standard CALViS pipeline YAML (Calvis_standard_pipeline_v1.0.yaml) into GPT before running a job for the first time:

1. Open the pipelines manager.



2. Import the Calvis_standard_pipeline YAML.



3. Confirm the imported pipeline is listed.



5. Run the pipeline on a grow

1. Select **New Job** and choose **Calvis_std** from the pipelines list. Choose your grow, name your gpro, and set the gpro output location.



Important

The gpro **must** be saved under **T1_proc**.

2. Configure GNSS processing. We currently recommend **PPRTX**.

General

GNSS

LIDAR

VNIR

SWIR

+ Add Script

GNSS: APX-15Lv3

Flight 1

Force Reprocess ☒

First POS File 6241C03151202604150124.T04 Select

Last POS File 6241C03151202604150133.T04 Select

GNSS Mode PP-RTX ▾

Datum GDA2020 ▾

Grid Australia (MGA2020) ▾

Zone MGA2020 MGA zone 54 ▾

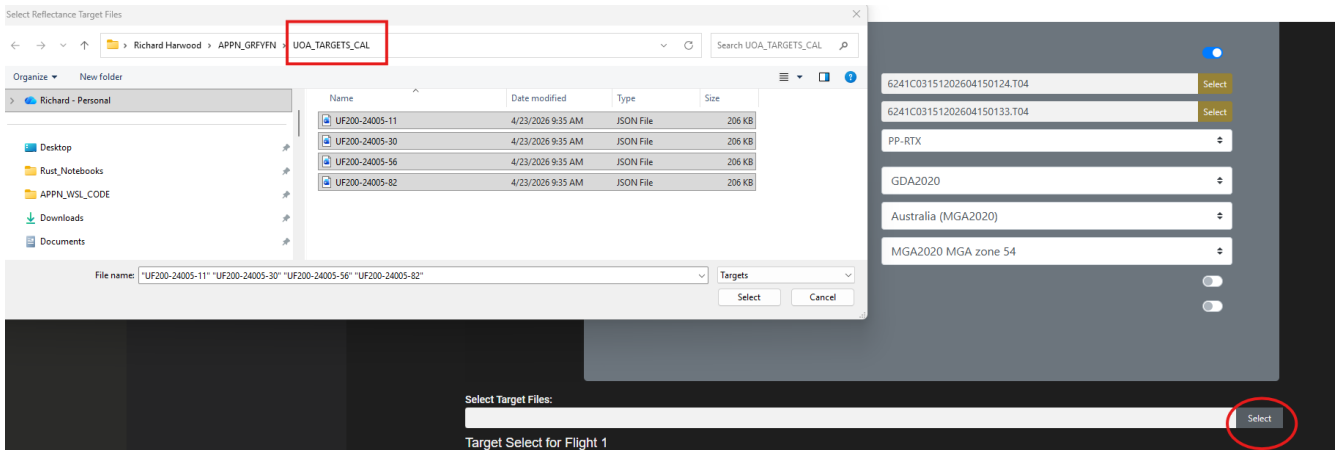
Realtime ☐

Override GNSS/IMU Lever Arms ☐

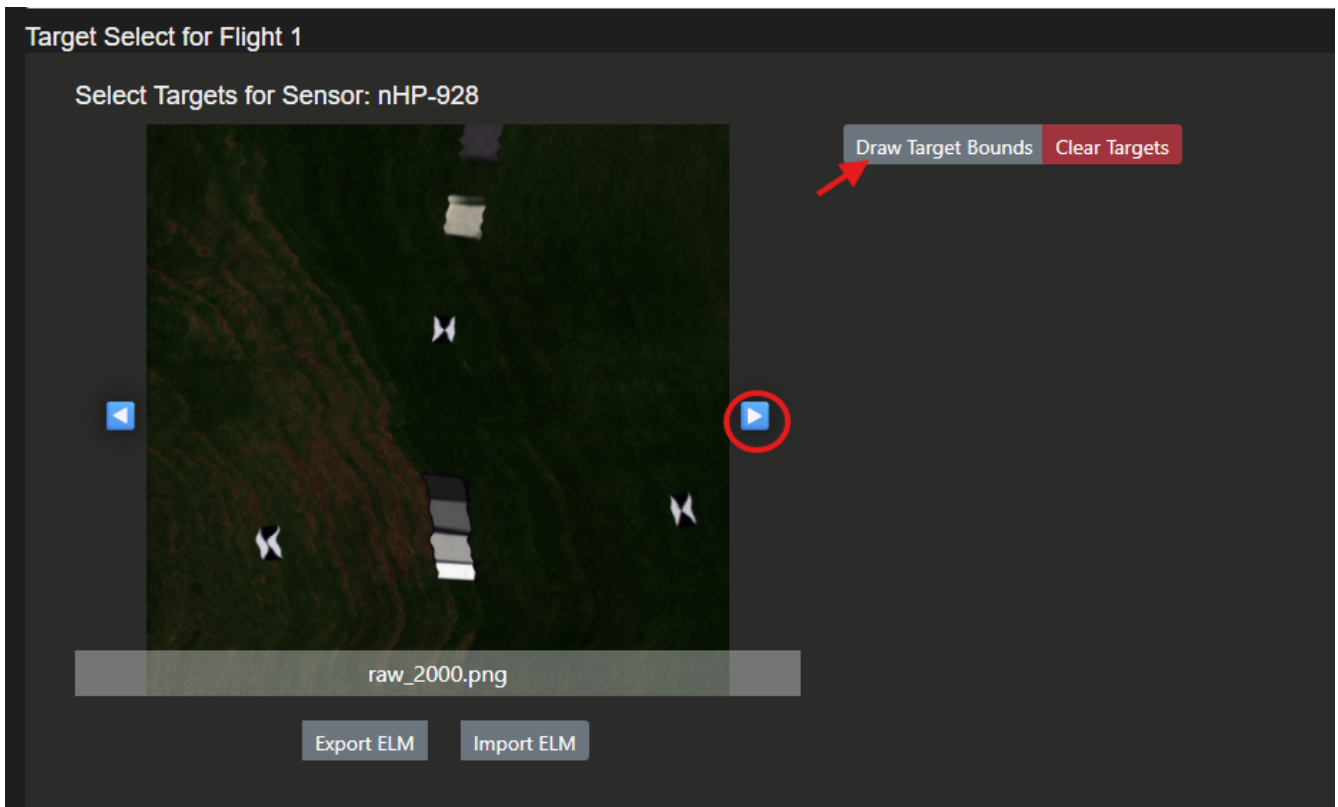
Caution

Double-check your **Datum**, **Grid**, and **Zone** before continuing. Errors here propagate through every downstream product.

3. Load the reflectance target files. For this example flight we use the UOA panel target files.



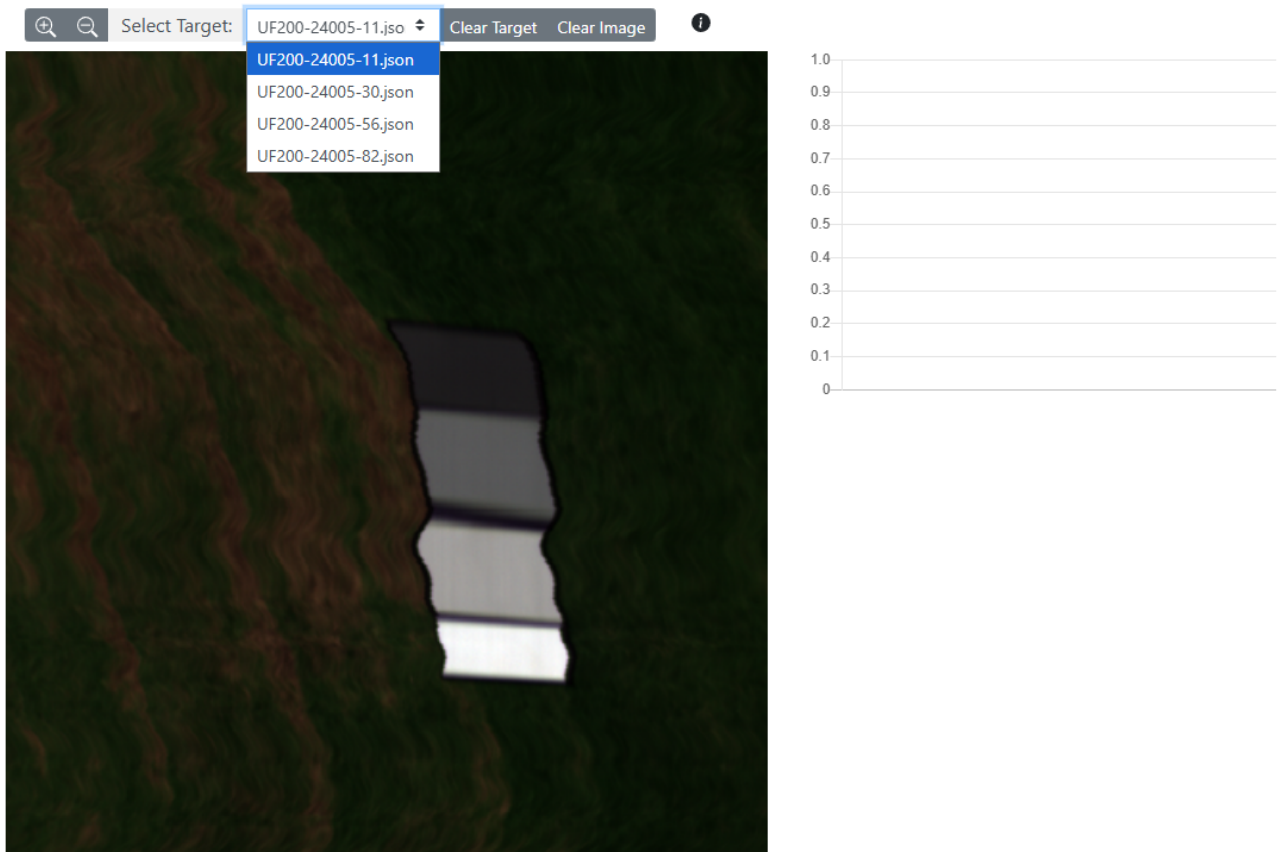
4. **VNIR ELM**. Cycle through the images until you find the four panels, then click **Draw target bounds**.



Choose a target to work on:

raw_2000.png ← →

Done



Hold right-click and drag a rectangle around the panel:

raw_2000.png

Done

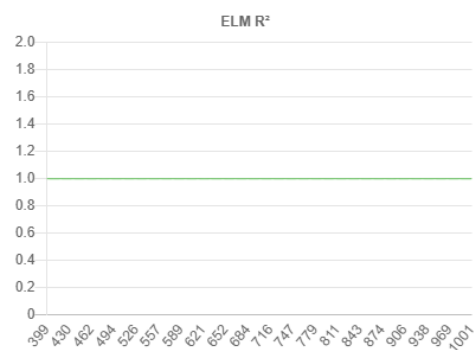
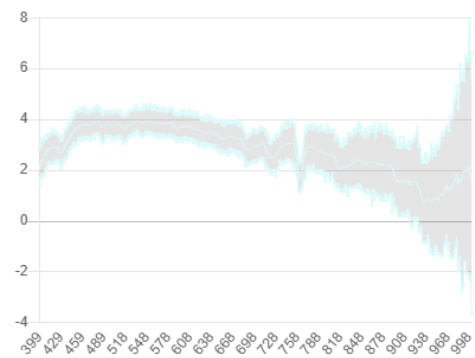


Select Target:

UF200-24005-11.iso

Clear Target

Clear Image

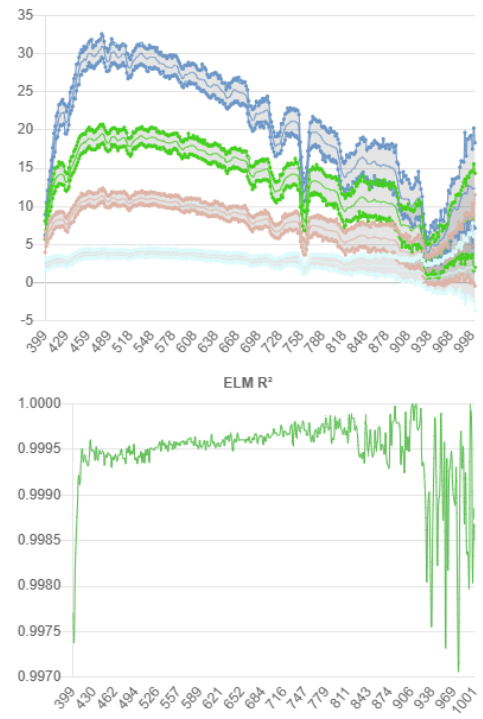


Repeat for the remaining three panels:

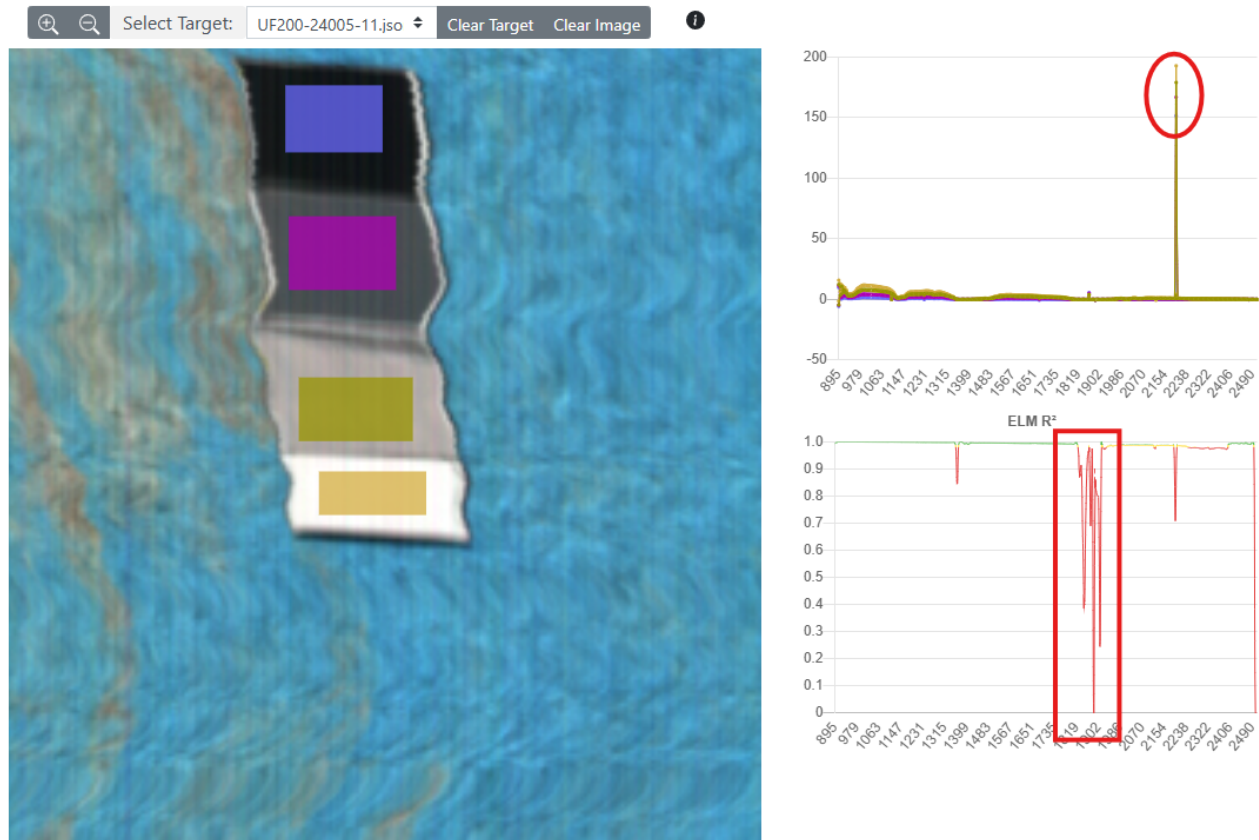
raw_2000.png

Done

Select Target: UF200-24005-82.jsa Clear Target Clear Image



5. **SWIR ELM.** The process is the same as VNIR.

**Tip**

The SWIR will have water bands (red box in the figure above) and sometimes rogue values (red circle). As a rule of thumb, **draw normal-sized boxes that cover the panel** rather than cherry-picking small areas to get a neater ELM.

6. Click **Submit**.

CALViS outputs

The CALViS standard pipeline produces the LiDAR and hyperspectral products listed below. See Standard Data Products for the canonical specifications, file sizes, and software compatibility.

Product	Output filename	Resolution	Format	Notes
LiDAR Digital Surface Model (DSM)	LiDAR_DSM.tif	8 cm (fixed)	GeoTIFF	Extent: VNIR scene
LiDAR Digital Terrain Model (DTM)	LiDAR_DTM.tif	1 m	GeoTIFF	Extent: processing extent
Combined LiDAR Point Cloud	LiDAR_CombinedPointCloud.las	Native point spacing	LAS	Outliers removed during combination
VNIR Orthomosaic	VNIR_Orthomosaic.bin	4 cm (GSD-based)	ENVI (.bin + .hdr)	binning = 2, radiometric calibration a
SWIR Orthomosaic	SWIR_Orthomosaic.bin	4 cm (GSD-based)	ENVI (.bin + .hdr)	binning = 2, radiometric calibration a

GOBI — Standard Processing Pipeline

The GOBI standard processing pipeline is defined by `Gobi_standard_pipeline_v1.0.yaml`.

Note

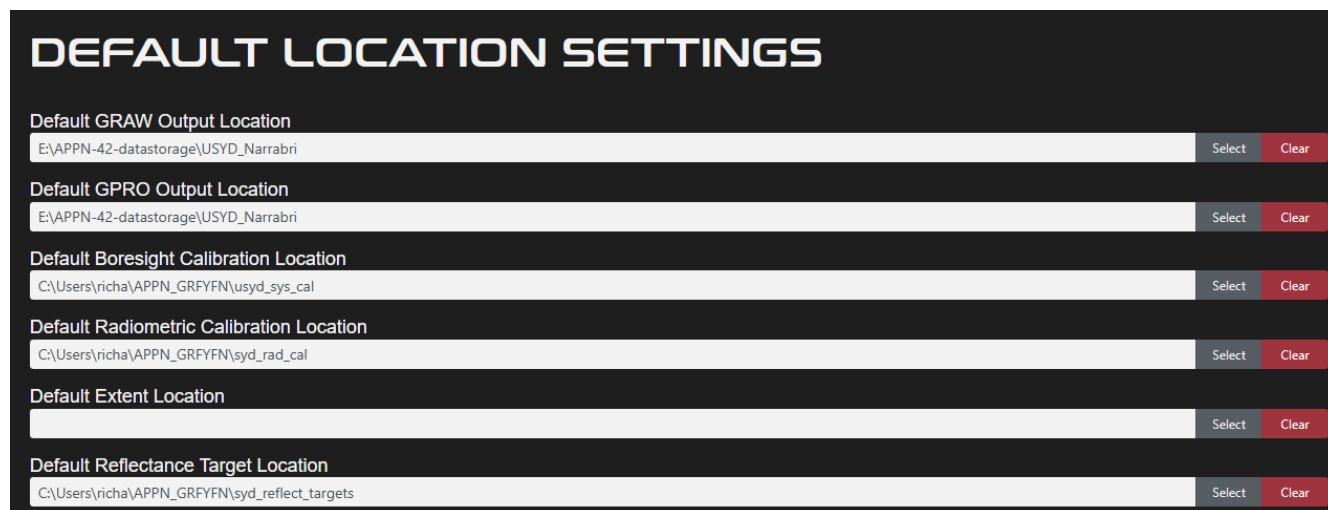
The GOBI standard processing walkthrough is **TODO**. The outputs of the standard GOBI pipeline are documented below.

The GOBI processing workflow is very similar to the CALViS. The main difference is that there is no SWIR but there is RGB data.

1. Configure GPT defaults

Repeated to make sure double checking is done

Before processing any data, configure GPT so that it uses the correct radiometric calibration files and points to the correct panel target files.



Setting	Value	Select	Clear
Default GRAW Output Location	E:\APPN-42-datastorage\USYD_Narrabri	Yes	Yes
Default GPRO Output Location	E:\APPN-42-datastorage\USYD_Narrabri	Yes	Yes
Default Boresight Calibration Location	C:\Users\richa\APPN_GRFYFN\usyd_sys_cal	Yes	Yes
Default Radiometric Calibration Location	C:\Users\richa\APPN_GRFYFN\syd_rad_cal	Yes	Yes
Default Extent Location		Yes	Yes
Default Reflectance Target Location	C:\Users\richa\APPN_GRFYFN\syd_reflect_targets	Yes	Yes

Sensor identification (boresight calibration) Each GOBI unit has a unique sensor identifier embedded in the boresight calibration filename. The naming convention is:

Example: 20241008_GOBI2405-APPN3_SystemCal.yml

Where APPN3 is the USYD sensor number; each GOBI unit is assigned its own unique number.

Important

When processing data, you **must** verify that the sensor number in the boresight calibration file matches the sensor used for data collection.

Radiometric calibration files Each sensor has an associated set of radiometric calibration files supplied by GRYFN.

Important

Verify that the **Radiometric Calibration Location** parameter in GPT points to the correct calibration files for your specific sensor(s).

Reflectance target values The Empirical Line Method (ELM) requires accurate reflectance target values for the calibration panels.

- Four (4) calibration panels are used for the ELM process.
- Each panel has specific, measured target reflectance values.
- These values are unique to your panel set.

Important

Ensure the **Reflectance Target Location** parameter references the correct target values that correspond to your specific calibration panels.

2. Format the raw data

The example CALViS flight used here is **run_00**, a flight from USYD.

A complete CALViS dataset has four key components — **VNIR, RGB, LiDAR, and GNSS** — and these must be arranged correctly before a working graw can be bundled. The raw data folder will look like this:

Example CALViS RAW folder layout showing VNIR, SWIR, LiDAR and GNSS components

3. Create the graw bundle

Pre-bundle checks Before each graw, confirm you have:

- ☐ System calibration file (matching the sensor used).
- ☐ Radiometric calibration files.
- ☐ Reflectance target file.

UOA calibration paths

Note here we switched from UOA to USYD

Bundling steps

1. **Set the system calibration file** to match the sensor you are using.

GPT system calibration selection dialog

2. **Choose the raw data path.**

GPT raw data path selection

3. **Click *Next*** to view the optional parameters.



Note

Unless you have a specific reason to set an extent, **leave it blank** — data can be cropped to the hyperspectral capture area later, which is ideal for most standard flights. Other fields here are also optional.

4. **Click *Next***. Give the bundle a logical name and save it under **T0_raw**.

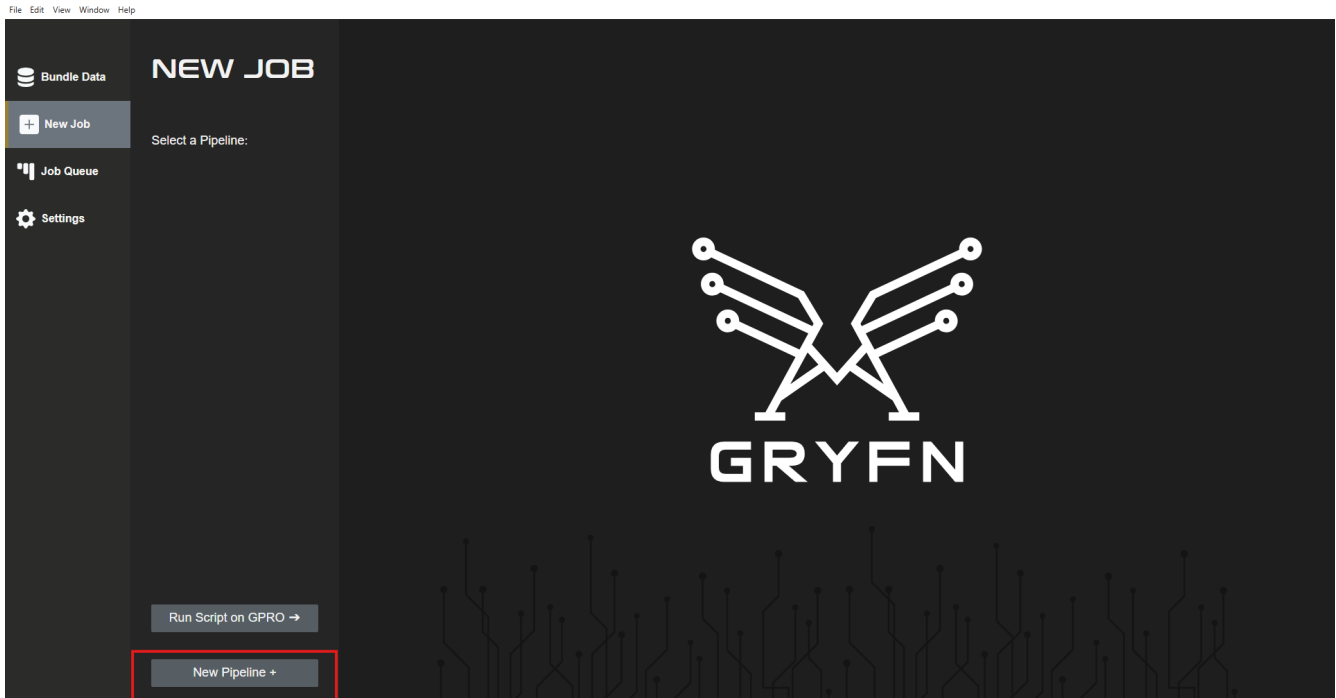
Save graw dialog with bundle named and saved under **T0_raw**

5. **Click *Create bundle***.

4. Load the GOBI_standard_pipeline

Import the standard GOBI pipeline YAML (Calvis_standard_pipeline_v1.0.yaml)

1. Open the pipelines manager.



2. Import the Gobi_standard_pipeline YAML.

GPT pipeline import dialog — step 2 of upload

3. Confirm the imported pipeline is listed.

GPT pipelines manager showing imported Calvis_standard_pipeline — step 3 of upload

5. Run the pipeline on a graw

1. Select **New Job** and choose **Gobi_std** from the pipelines list. Choose your graw, name your gpro, and set the gpro output location.

New job dialog: graw selection and gpro output path

Important

The gpro **must** be saved under T1_proc.

2. Configure GNSS processing.

GNSS processing configuration dialog

Caution

Double-check your **Datum**, **Grid**, and **Zone** before continuing. Errors here propagate through every downstream product.

There is no PPRTX option for GOBI.

3. Load the reflectance target files. For this example flight we use the UOA panel target files.

Reflectance target file selection — USYD panels

4. **VNIR ELM.** Cycle through the images until you find the four panels, then click **Draw target bounds**.
(note explained in greater detail in the CALViS steps above)

Selecting a VNIR ELM target panel

GOBI outputs

The GOBI standard pipeline produces the LiDAR, hyperspectral, and RGB products listed below. See Standard Data Products for the canonical specifications, file sizes, and software compatibility.

Product	Output filename	Resolution	Format	Notes
LiDAR Digital Surface Model (DSM)	<code>LiDAR_DSM.tif</code>	8 cm (fixed)	GeoTIFF	Extent: VNIR scene
LiDAR Digital Terrain Model (DTM)	<code>LiDAR_DTM.tif</code>	1 m	GeoTIFF	Extent: processing extent
Combined LiDAR Point Cloud	<code>LiDAR_CombinedPointCloud.las</code>	Native point spacing	LAS	Outliers removed during combination
VNIR Orthomosaic	<code>VNIR_Orthomosaic.bin</code>	4 cm (GSD-based)	ENVI (.bin + .hdr)	binning = 2, radiometric calibration applied
RGB Orthomosaic	<code>RGB_Orthomosaic.tif</code>	0.6 cm (fixed)	GeoTIFF	Feature-matching (SIFT) bundle adjustment